

COKER

IMPROVISING JAZZ

9-88 \$1.95



IMPROVISING JAZZ

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88



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Introduction

Few methods have been designed for group improvisation, which, in essence, is a spontaneous exchange or interplay of musical ideas and moods. Jazz music, with its roots in basic rhythms and simple melodies, has developed *naturally* into a blend of musicianship, humanity, and intellect, having universal appeal. Improvisation has existed in other styles, but in the classical music of Western civilization its use has been stifled by enlarged instrumentations and the complexity of compositional techniques which have made no allowances for this means of individual expression. The composer achieves the effect of spontaneity when his written music flows naturally and is well played.

Jazz has brought about a renaissance in improvisation, providing a style which is conducive to spontaneous creation by utilizing standard musical elements, such as 4/4 time, songs of uniform length and form (usually 32 measures in length, with an A-A-B-A structure), fairly standardized instrumentation, steady tempi, consistent and logical harmonies, stylized melodies and rhythms, and even an established order of introductions, statements of themes, sequence of soloists, and codas and endings. Such an established framework as we find in jazz improvisation is as useful to the jazz player as the twelve tone system is to the atonalist composer. The characteristics of the style make for swift decisions, enabling the music to move

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along without interruption. This is not to say that jazz music has always been and will always stay within the aforementioned boundaries dictated by its style, any more than it would be correct to state that twelve tone music is without potential for further development. Jazz has already begun to expand its resources by absorbing the multitude of musical techniques existing in other styles of music. Improvisation in any of the existing styles offers the musician the opportunity to utilize his technical ability to its fullest extent, while enjoying the creative freedom of spontaneous composition. In our present culture, the bulk of the activity in improvisation is in jazz music.

This book is designed to equip you with the understanding of the theoretical principles used in jazz, presented in logical sequence as they apply to the ultimate improvised performance. As a prerequisite to this study, the student must have some technical proficiency and should be reasonably acquainted with major and minor scales.

The absorption and utilization of theory and techniques, which are the improvisor's tools, can in no way guarantee an interesting musical personality. Each jazz player will find his own musical style and his style will be subject to the criticism of the listener. He may develop a style which is physical or cerebral, faddish or original, blatant or subtle, rambling or formful, ugly or beautiful, flexible or inflexible, tense or relaxed, exciting or dull, or any combination of these extremes, all of which could be the product of a technically equipped player. The style of the individual player is affected by his personality, his intelligence, his talent, and his coordination, all of which are beyond the scope of this text.



the Improvisor's Basic Tools

Five factors are chiefly responsible for the outcome of the jazz player's improvisation: intuition, intellect, emotion, sense of pitch, and habit. His intuition is responsible for the bulk of his originality; his emotion determines the mood; his intellect helps him to plan the technical problems and, with intuition, to develop the melodic form; his sense of pitch transforms heard or imagined pitches into letter names and fingerings; his playing habits enable his fingers to quickly find certain established pitch patterns. Four of these elements of his thinking—intuition, emotion, sense of pitch, and habit—are largely subconscious. Consequently, any control over his improvisation must originate in the intellect. While the intellect is limited in its capacity for control over intuition and emotion, it can be responsible for the training of the ear and for establishing a variety of helpful finger patterns, in addition to its function of solving technical problems.

It would be difficult to place these five factors into proportionate values. Some improvisors rely more heavily on certain factors; others will depend on other factors. A few gifted players are able to perform adequately by relying completely on the subconscious elements. All but the rare genius, however, are eventually limited in their development. They need special study, due to the problems of deeply ingrained habits, the unaccustomed rigors of work-

up to every potential, and, in some instances, an inability to it or evaluate shortcomings. The ability to evaluate is important and will be discussed in Appendix A.

Since the intellect is the only completely controllable factor, we approach the problem of learning to play jazz almost solely through this factor, and hope that the other four (intuition, emotion, the sense of pitch, and habit) will progress at the rate established by the intellect. If, in the first lessons, the approach seems uncalculated, remember that most artistic accomplishment requires academic training. This training is the foundation upon which to build, and it will strengthen your capacity to enjoy fully your own work and the work of others.

Let us now look more deeply into the problems of the intellect. An improviser must know, for his own musical security, the general framework on which he bases his improvisation. This, in most cases, is a song form, either twelve or thirty-two measures in length, which is repeated as many times as necessary so that maximum melodic freedom is allowed to each of the individual improvisors in the jazz ensemble. The player's knowledge of the material must include: (1) the length of the tune; (2) its thematic and harmonic construction, in a general sense (A-B-A, A-A-B-A, etc.), and the length of these sections; (3) the tonality of the tune and any temporary modulations to other keys; (4) the individual chords of the progression and how they are related to one another; (5) the scales which fit the various chords and sections of the tune; and (6) the emotional equality or mood of the tune.

These are by no means the *only* considerations confronting the improviser. Rather, they comprise the basic minimum of his needed working knowledge of the material. Additional considerations will be discussed after you have grasped the more general techniques of improvising.

Let us examine and analyze, in the light of the previously mentioned aspects, an example of a typical tune used by improvisors, the "blues" tune. (The terms *tune* and *song* are used in a liberal sense here, since the original melody is largely ignored and obscured by the jazz player's improvisations. His creative endeavor is ended by his own melodies, based on the *chord progressions* of the

tunes he uses.) The blues progression shown in Figure 1 is only one of many existing chord progressions to that tune and is an example of one of the more basic and simple patterns.

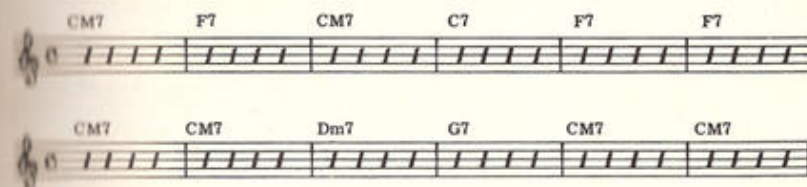


FIGURE 1

The slanted lines below each chord symbol represent the beats. The first letter of each of the chord symbols indicates the pitch upon which the chord is built, called the *root*. A capital "M" signifies a *major triad* and a small "m" indicates a *minor triad*. A triad is a three-note chord made up of the first, third, and fifth degrees of a scale, a major scale for a major triad and a minor scale for a minor triad. The number "7" indicates that the seventh degree above the root has been added to the triad, transforming it into a *seventh chord*. If the symbol for the seventh contains a capital "M," the chord is constructed from the first, third, fifth, and seventh degrees of a major scale built on the root of the chord. A natural minor scale is used to determine the correct spelling of the first, third, fifth, and seventh scale degrees of a chord having the symbol, "m7." If the symbol includes only the root name and a "7," without the "M" or "m" to indicate whether the chord is major or minor, it is understood that the triad is major, but the seventh is lowered a half-step from its position in the major scale. The written names for these types of seventh chords are:

M7—major seventh chord

m7—minor seventh chord

7—seventh chord or dominant seventh chord

Examples of the construction of these chords would be as follows:



A clearer projection of the harmony of the blues progression of Figure 1 would now be:



In further analyzing the blues, we find the tune to be twelve measures long, and in the key of C major. (For the present, let it suffice to say that we guess it to be in C major because it starts and ends on a C major seventh chord.) Since there is no given melody in Figure 1, we cannot determine the motivic construction; however, notice that the progression seems basically to contain four measures in C, then an implied feeling of F for two measures, followed by six measures of C chords and chords which are closely related to the key of C. These sections might be labeled A-B-A. We will discuss key modulations in a later section, when you have digested the terminology and information needed for such analysis. The term *blues*, in jazz, usually denotes a chord progression twelve bars in length, and also describes its mood.

Since melody ordinarily moves by steps, rather than from one chord tone to another, it would be helpful to decide what notes can be played *between* chord tones. For the major seventh chord

we shall use a major scale built on the root of the chord; hence the scale for the CM7 chord would be a C major scale. This gives us the additional notes D, F, and A to act as melodic joiners between the more important melodic notes on C, E, G, and B. For the F7 chord we will use an F scale, but lowering the seventh degree (E) by a half-step to accommodate the Eb used in the chord.



The Dm7 and G7 chords, as stated earlier, are closely related to the key of C. Therefore a C major scale may be used on these chords, but with the scale beginning on the roots of those chords.



The above scale on D is an example of a scale construction known as the *Dorian Mode* which belongs to a family of modal scales discussed in music theory and music history texts. The scale on G, also shown above, corresponds to the construction of another modal scale from that family, the *Mixolydian Mode*. Hereafter, the scale which uses the same tones as a major scale of a major second (two half-steps) lower will be referred to as the *Dorian Mode*. Similarly, the scale whose root is the fifth degree of a major scale, and which uses the same pitches, will be called the *Mixolydian Mode*.

Figure 2 shows a summation of what we have just learned about the blues progression. This information should be analyzed and studied carefully as preparation for improvisation on the blues progression of Figure 1.



2

an Introduction to Melody

Melody is (1) one of the essential elements of music, along with harmony and rhythm; (2) that part of music which is heard most prominently; (3) a component of music capable of division into smaller fragments, such as periods, phrases, or motifs; and (4) a group of fragments woven into symmetrical patterns. The *motif* is the smallest melodic entity from which much of the remainder of the music is written or played.

Figure 3 shows a twelve-measure solo on the blues progression,



FIGURE 3

having six melodic fragments (in brackets) which occur in a symmetrical pattern. The last five fragments are related by obvious similarities to the first fragment, which we shall call the *motif*. Because they vary only slightly from the motif, these five fragments would be called *variations*. The use of this sort of melodic organization, often called the *theme* (motif) and *variations* form, seems to be a natural tendency of both improvisors and composers. Most music, whether composed spontaneously or not, can be analyzed, therefore, in terms of motifs and variations of motifs. The degree to which this type of melodic form exists would of course depend upon the individual artist's ability and desire to use it.

Some jazz solos are made up of a steady stream of eighth notes, divided into unrelated phrases by infrequent pauses for breath or thought. We shall call this type of improvisation a *linear* style. It is sometimes difficult to analyze in terms of motivic construction, but we can appreciate it for its continuity, smoothness, contours, note choices, and placement of rhythmic accents, rather than for its melodic form. Figure 4 is an example of a solo in the linear style.



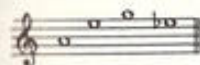
FIGURE 4

Linear melodies may or may not possess formal symmetries, and motivic melodies exist in which there are no variations or repetitions. These are called *through-composed melodies* and may result through either contrivance or accident. Communication with the listener is reduced by through-composed melodies and, though they are sometimes effective, avoid them until you develop melodic

form and smoothly coordinated phrases through the study of the theme and variations technique.

Just as the young composer learns much about his craft by listening to music while following the score, the beginning improviser can progress more rapidly by reading a transcription of an improvised solo while listening to the recording. If no transcriptions are available, you will have to transcribe them yourself, beginning with relatively uncomplicated solos and gradually trying more difficult ones. We cannot emphasize this practice too much; it will benefit you in two important areas: (1) it will develop your ear and pitch memory to the extent that you will eventually be able to transcribe *your own* ideas while you are improvising; and (2) by studying the solos and styles of already proficient improvisors, you will gain a deeper understanding of the improvised solo and will discover various methods and ideas for the handling of improvised material. Your sense for evaluating the merits of various soloists will also increase as you discover which solos can bear the careful scrutiny of analysis. Once the solo has been transcribed as accurately as possible, try to locate motifs, variations of motifs, and noteworthy linear sections, as shown in Figures 3 and 4.

In addition to studying solos through the use of transcriptions, it would be wise to begin an orderly and faithful collection of *original* motifs to be used for analysis and development. These will eventually become part of your jazz style and part of your repertoire of original ideas to be used in future improvisations. By studying Figure 3, you should get an understanding of the general appearance and nature of a motif. It is a short but complete melodic idea, usually having a logical rhythmic close, either by the use of a note of greater duration or by the use of rests. Anyone, even the beginner, is capable of composing short, original motifs, and with a little practice can become increasingly fast and accurate with putting them on paper. One or two motifs per day is reasonable as a beginning minimum. The output will increase naturally. The motifs should be neatly entered into a notebook (open to revision) and used as a source book of musical ideas. If you have difficulty composing motifs, you might begin by finding several pitches which sound good in succession and which seem to be relatively original, writing them as whole notes for the moment and adding a rhythm to them later.



The ideas need not be striking or unusual. Don't wait for what you consider an exceptional motif to come to mind; even mildly effective ideas can become richer through thoughtful development, reworking, and variations. Simple, uncomplicated motifs are often more effective than more complicated ones. Furthermore, the ideas need not depict the style of "bluesy" jazz, with an abundance of lowered thirds, lowered fifths, and lowered sevenths, the ingredients often found in what is commonly termed a "funky" idea. The "funky" style will develop quite naturally in appropriate spots without encouragement, because of its popular usage and its suitability to the jazz style.

Successive, or at least well-placed, variations of motifs, when employed by the jazz improviser, have a definite, strengthening effect upon the relationship between the performer and the listener. Richmond Browne, jazz pianist and instructor of theory at Yale University, wrote in a letter to the author:

What is the soloist doing when he attempts to "build"? Actually the ideal process hardly ever takes place—that is, it is hardly ever the case that a conscientious soloist plays a thinking solo for a hard-listening hearer—but when this does happen, the key process is memory. The soloist has to establish for the listener what the important POINT, the motif if you like, is, and then show as much as he can of what it is that he sees in that motif, extending the relationships of it to the basic while never giving the feeling he has forgotten it. In other words, I believe that it should be a basic principle to use repetition, rather than variety—but not too much. The listener is constantly making predictions; actual infinitesimal predictions as to whether the next event will be a repetition of something, or something different. The player is constantly either confirming or denying these predictions in the listener's mind. As nearly as we can tell (Kraehenbuehl at Yale and I), the listener must come out right about 50% of the time—if he is too successful in predicting, he will be bored; if he is too unsuccessful, he will give up and call the music "disorganized."

Thus if the player starts a repetitive pattern, the listener's attention drops away as soon as he has successfully predicted that it is going to continue. Then, if the thing keeps going, the attention curve comes back up, and the listener becomes interested in just how long the pattern is going to continue. Similarly, if the player never repeats anything, no matter how tremendous an imagination he has, the listener will decide that the game is not worth playing, that he is not going to be able to make *any* predictions right, and also stops listening. Too much difference is sameness: boring. Too much sameness is boring—but also different once in a while.

Most improvisors and composers have a natural inclination to use melodic form, but often to an insufficient degree. Both intellect and intuition must come into play if the solo is to be wholly satisfying. Methods for developing motifs should be studied, written, and played. If your original motif collection is used as a source for melodic development, it will follow that *all* the material developed from those motifs will also be original, and you will automatically develop an original style.

The only developmental method we will be concerned with in this chapter is *transposition*, which is simply a matter of rewriting a motif so that it fits another key, chord, or type of chord. Let us say that you have composed an effective motif which seems to fit nicely into the key of C. This means that the motif will serve you well, *as long as the piece begins and remains in C*. But of course not all tunes are likely to be played in the key of C, under normal circumstances; indeed, most tunes contain *at least* one simple modulation to another key. You certainly cannot afford to wait for a certain key, chord root, and chord type to use your motif. Furthermore, the continuity would be broken if you use it only at those widely spaced intervals. Therefore it would be well to learn to write and play your motifs in all keys (by transposition) and against all types of chords (by transposition and minor alterations, which will not affect the general character of the idea). Also, you will discover that many of your motifs can begin on another member of the accompanying or implied chord. To know what chord fits with a motif can seem like an overwhelming problem to the beginning

student of improvisation. However, common sense and taste are the only guides to finding the "correct" chords; rarely can a motif be harmonized in one way only. This explains why a multitude of different chord progressions can (and do) exist on a single given tune when played by different improvisors of various styles.

To find an arbitrary chord which will fit the motifs, rearrange (if necessary) the tones used, being especially careful to include the important tones of longer duration (you will find that some can be omitted as embellishing, nonessential tones) so that they are stacked in third intervals as much as possible, simulating the construction of a triad or seventh chord. (Ninths, elevenths, thirteenth, and altered chords can be added when they become part of your harmonic knowledge.) Then analyze the resulting chord or chord fragment to determine the root and type of chord which should be used to accompany the motif.



Once the probable chord (or chords) has been determined, the motif is ready for transposition. Since the above motif is harmonized by an Em7, the first step would be to transpose it, raising it a half step each time (Fm7, F#m7, Gm7, Ab m7).



Now suppose, after having all the transpositions of the motif, you are confronted with an E7 rather than an Em7 chord. The motif, in order to fit a different type of chord, will probably require some alteration. There are many ways to vary the tones in the idea, without really changing its character. One way would be:



Regardless of the type of chord or the particular pitch of the root of the chord, the motif will adapt readily in a number of ways.

To add another dimension, it is usually possible to place the motif in a different area of the same chord or of a different chord. The motif under discussion begins on the root of the Em7 chord, but it could be revised to begin on the third (or fifth, seventh, etc.) of an Em7 chord (or any other chord or chord type, for that matter).



Once you have learned the motif in twelve keys, with M7, m7, and 7 chords, and built on at least two other members of the chord, you will have increased the number of useful circumstances for that motif (and some of these will offer interesting variations) from one situation to *one hundred and eight situations!* The transpositional possibilities for development are awesome and so may seem tedious to write; however you will find that you will soon be able to transpose quickly enough to dispense with writing them and simply play them in exercise form as preparation for their use in actual improvisation. An example of how these transpositions might appear in a solo would be:



Keep in mind that melodic variations have at least two objectives, both of which will raise the level of communication between the performer and the listener: that of offering contrast to the original version of the melody, and that of maintaining contact with the motif. By offering contrast, the variation extends the improvisation

without risking boredom, and by having similarities to the original motif, the variation affords structural unity to the improvisation.

projects

1. Begin transcribing of solos, as suggested in this chapter.
2. Begin collecting of original motifs.
3. Learn to develop melodies by transposition, using the motif collection as a source of material.

3

the Rhythm Section

The rhythm section, because it provides the pulse, meter, rhythmic accents, compassionate or sympathetic balance, rhythmic "swing," and harmony, constitutes the heart of an improvising group. Some small jazz groups contain *only* rhythm section players—combinations of piano, bass, drums, and sometimes guitar, vibraphone, accordion, or Latin American instruments, such as conga drums, timbales, bongos, maracas, claves, guiro, cowbell, etc. In these cases the rhythm section must also supply *all* of the improvised solos and the foreground duties usually assigned to the wind instruments. In an optimum situation, when the members of the rhythm section are reasonably accomplished, their function is not simply to provide a throbbing background to a featured wind instrument, nor to play together in rhythmic and harmonic unison, as an indissoluble group. Rather they share the burden of the wind instrument by serving as a soundboard for its rhythmic accents, harmonic deviations, melodic continuity, and mood changes. They contribute more than their basic duties of time-keeping and background vamping by feeding the soloist with new material to be developed coordinately. At times they can even become cooperative entities within the rhythm section, echoing each other, dropping out occasionally, or becoming part of the foreground without necessarily involving the other members of the

section. We shall limit our study to the basic foundations of rhythm section playing, as applied to the most common instrumentation of that section, the piano, bass, and drums.

drums

Contrary to common belief, the modern drummer does not rely upon the rhythmic figures of the bass drum or snare drum for maintaining the pulse (beat). He does, however, need to use some component of his drums for beating steady, dependable time, to which he can coordinate the improvised rhythms of the other components of his set. Oddly enough, the modern drummer, perhaps for the sake of subtlety, uses his cymbals for this purpose, either the "ride" cymbal (a large, medium thin, mounted cymbal), the "hi-hat" ("sock" cymbals), or both, played with either sticks or brushes. Although the patterns played on these cymbals can vary with the individual, the most common rhythm is:



If brushes are used, the ride cymbal pattern is often played on the snare drum with one hand, while the other hand rotates a brush around the edge of the snare head, making a "swishing" sound accenting the second and fourth beat by accelerating the swish sharply toward the body. The brushes can also be used on the cymbal, however.

The bass drum, snare drum, and tom-toms (if they are used), are usually reserved for improvised accents and this is left to the drummer's taste and discretion. Many drummers like to beat their bass drum lightly, even inaudibly, in steady quarter notes except when it is being used for accents. The pulse will be supplied by the string bass player, who will generally resent a heavily sounded bass drum which obscures the sound of the bass. In the unfortunate absence of a bass, a heavier bass drum is sometimes helpful, but it is unsatisfactory because of the short duration of its sound. Each of the pizzicato bass tones should have a long, singing sound.

bass

The rhythmic pattern of the bass is quite simple, usually quarter notes with occasional eighth-note figures to break the monotony. Once in a while the bass will play *in two* (quarter notes on the first and third beats) for a section of the tune, occasionally using quarter notes on second and fourth beats for variety.

As for the pitches played by the bass, there are several types of bass lines. The bass player's first obligation is to outline the consonant and important notes of each of the chords (especially roots and fifths), or he should at least outline the important notes of the tonality. The two most prominent types of bass lines can best be described as *chordal* and *scale-like* ("walking" line). The chordal line is the easier of the two and is recommended for beginners. The notes are chosen from the given chord, with emphasis given to the roots and the fifths. An example of such a line would be:



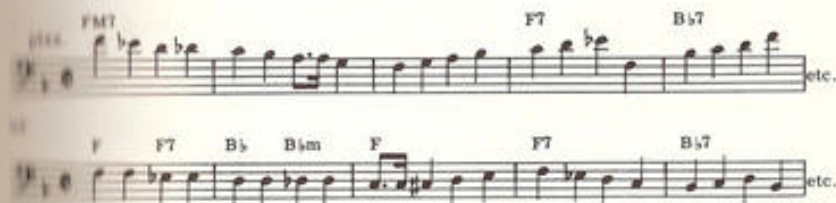
Note that the line contains practically no sevenths, but is more inclined to outline the triad portion of the chords. Because of the low range of the bass, the dissonance needs to be reduced by comparison with higher-pitched instruments. Simplicity and economy is of utmost importance for bass players, especially at the beginning. The line above would sound nearly as good even with successive repetitions of some notes. For example, the first measure could have been F, F, C, F, or F, F, C, C, instead of F, A, C, A (see example).



This is left to the taste, ingenuity, and technical ability of the player. The second and fourth beats can contain non-chord (dissonant) tones if the player so desires. For instance, the third measure could have been C, B-flat, A, F, rather than C, F, A, C. It could

even be C, B-flat, A, G, since the non-chord tones would still be on the second and fourth beats. Traditional theory teaches us that both the metric and melodic accents in a quadruple meter (i.e. 4/4) occur on the first and third beats. This leaves the second and fourth beats unaccented, and therefore a logical place to use the nonharmonic tones of a melody. In jazz, the method for handling nonharmonic tones is essentially the same as in classical theory, but the metric accents are placed on the second and fourth beats. Consequently, the bass player should accent the second and fourth beats even though they might be dissonant (nonharmonic) tones. The first and third beats, of course, will be unaccented and consonant. It is best, when inserting non-chord tones, to choose tones within the scale of the key, hence the B-flat instead of B.

The walking bass line is scale-like, using considerably more non-chord tones and even chromatic tones (tones not within the key signature). The walking line would look something like the following:



piano

The piano player is the one member of the rhythm section who has no definite rhythm pattern to maintain, yet his improvised rhythmic figures add greatly to the time-keeping. Perhaps the beginner should play in steady whole notes, half notes, or quarter notes, trying gradually to invent mixtures of these and to enhance them with syncopated figures. He should listen closely to the drummer for rhythmic figures which he can adapt. The soloing wind instrument may occasionally feed him possibilities also. The piano player should not be afraid to repeat figures once in a while, in a vamping style, changing the chords when necessary, while

retaining the rhythmic figure. The pianist carries on three functions simultaneously: (1) the rhythm, including keeping tempo, maintaining the meter, and inventing interesting rhythmic "punches"; (2) the harmony; and (3) the element of melodic imitation, derived from the soloing instrument. He will soon learn, as the drummer does, that certain rhythmic figures will fit the meter, others will not, and experimentation will be his chief means of finding out.

He can, however, be prepared to play a variety of harmonic voicings by practicing them diligently, so that when the group assembles to play, his mind will be freer to concentrate on rhythmic figures and time-keeping. There are no set rules as to how he must arrange (voice) the tones of each of the chords; however there are certain tendencies established by taste and tradition which may be of help: (1) be flexible with voicings and strive for a variety; (2) try to avoid skipping by wide leaps when moving from chord to chord; (3) attempt, occasionally, to maintain the same top note, whenever it is common to several different chords, for at least several successive chords (all three of these points will encourage smoothness); (4) generally voice the root in the bass of each chord and space the chord wider near the bottom and closer near the top; and (5) use voicings of varying weight, depending upon the volume needed, playing fewer notes in softer passages and fuller-voiced chords in heavier passages.

Below are some suggested voicings for beginning pianists. Remember that these are suggestions for spacing and doubling that will apply to *any* type of chord, not just those shown here. These voicings should be practiced in all keys, on all types of chords. New ones should be invented and added whenever possible.

Voicing Code:	FM7	FM6	FM7	FM7	FM7
7th	5	6	5	5	5
5th	4	5	4	4	4
3rd	3	3	3	3	3
Root	2	2	2	2	2
7th					
Root		R	R	R	R

projects

Drums. Practice alone, following the suggestions for the cymbal and hi-hat figures. Play *only* the cymbal and hi-hat at first, until this feels natural and until the tempo remains steady. Be sure to accent the second and fourth beats. Then try using brushes in either of the prescribed manners. When these time-keeping practices become easy and natural, and approach a subconscious level, try improvising accents with the snare and bass drum (letting them imitate each other and occasionally having them in unison) *but do not alter the cymbal and hi-hat figures while practicing improvised accents* and be sure the tempo remains steady. Practice various tempos like these.

EXAMPLES OF SOME POSSIBLE RHYTHMIC FIGURES FOR SNARE AND BASS DRUM

- (1)
- (2)
- (3)
- (4)

Horn. Using the blues chord progression of Chapter 1, write several twelve-measure bass lines of both the chordal and walking type and practice playing them, striving for long pizzicato tones and a steady tempo. Transpose the lines to other keys and practice those. Practice various tempos.

Piano. Using the blues progression of Chapter 1, write and practice several twelve-measure voicings, keeping the type of voicing as consistent as possible. Play them in whole notes and half notes only, until a sense of meter develops, then try inventing various rhythmic figures which will fit the meter and which can be repeated for twelve measures.

4

the First Playing Session

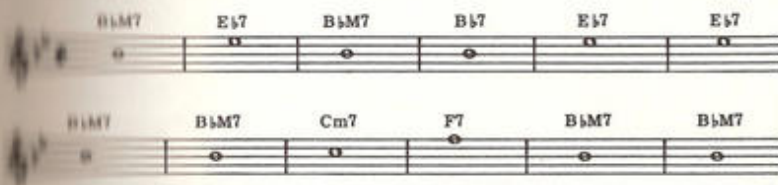
Before assembling for the first playing session, each participant should prepare himself thoroughly by studying the material to be improvised. It should be memorized, even though a reference sheet will be placed before him. Each player should also practice the material diligently by inventing arpeggio and scale patterns for all of the chord symbols, covering the entire range of the instrument. If every person will study the material carefully *before* each playing session, then his mind and fingers will not be entirely taken up with the details of knowing what notes are in each chord, what scale is used with each chord, and how the phrases should be fingered. The mind and fingers will be free to concentrate on more important aspects of playing, such as establishing melodic form; developing the meter sense, mood, and swing; finding useful notes and phrases; and planning and controlling the intensity of a solo in chosen or inspired moments. None of these elements can materialize if the mind is struggling with chord and scale fundamentals. As the player develops, he will find that many of these aspects will become relatively subconscious and easy. Now he will be able to add even more objectives.

The reference sheet should look like Figure 2 (Chapter 1), except that it may need to be transposed to fit the keys of the variously pitched instruments. (For the first playing session, we will use the

key of concert B $\flat$, which will place B $\flat$ -pitched instruments in the key of C, and E $\flat$ -pitched instruments in the key of G.) The chord progression will be the same as in Figure 2, except for transposition. The reference sheet should be placed before each player in succeeding sessions until he feels confident that it is no longer necessary.

The members of the rhythm section should be seated as closely together as possible, and the wind instruments should be seated in a semi-circle, near the piano. The rhythm section should then play through the progression a few times, while the others watch their reference sheets closely, adjusting to the feeling of the tempo, learning to feel the durations of each of the chords, listening to the differences between successive chords, and trying to maintain the sound of the keynote in their minds. The pianist should play as simply and rhythmically as possible, sometimes playing whole notes on the chords.

Next, the others should join in, playing several choruses of whole notes in unison, and playing only the roots of each chord.



This will train the player to hear the foundation of the progression and will help develop his sense of meter as he plays the harmonic rhythm (the relative duration of each chord). In this first session the harmonic rhythm is quite simple, since none of the chords has a duration shorter than a whole note.

The wind instruments should then be assigned a certain note to each chord to play for one chorus. For example, one could play the thirds of all the chords (in tempo), so that he would play (in concert key here) the following pitches in whole notes: D (third of B $\flat$ M $\flat$), G (third of E $\flat$ 7), D, D, G, G, D, D, E $\flat$, A, D, D. Another player would play all the fifths, another the sevenths, and another the roots. It won't matter too much which octave they choose for playing the assigned pitches, although it will sound best if the roots and fifth

4

the First Playing Session

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are in a lower range than the thirds and sevenths. After one or two choruses of this, exchange notes until each player has had the opportunity to play all the members of the chords. This exercise will strengthen the ear in understanding the sound of each of the chords, and will also teach you to think more quickly in terms of chord spellings.

Next, the wind instruments should play the following arpeggiated figure for several choruses (in unison):



This exercise begins to train the fingers to find correct pitches, though the pattern is repetitious. As explained in Chapter 1, habit plays an important role in improvising. By practicing exercises of this type, plus more of your own invention, you will form a foundation of correct finger habits. It is best that this type of exercise, as well as those described earlier, be practiced without reading, so that the intellect is encouraged toward deeper concentration.* Another exercise would be:



Eventually, when the progressions begin to move a little faster and half-note durations occur on some chords, such a pattern will have

* The development of concentration is a necessity for the improviser. Not only will the player progress rapidly through complete concentration, but he will also learn to combat self-consciousness in performance. The accomplished improviser has usually developed his powers of concentration to a very high degree. Mood and continuity are more consistently sustained when the player is not easily distracted. Also, a more technical and interesting performance is possible. By giving his attention wholeheartedly to the best possible contribution, the improviser can relax quickly and dispense with the unnecessary physical effort that the rattled player must make. Advanced or gifted players may rely solely on emotion for total involvement. This is concentration of a type which transcends early technical training, and which eventually displays the musical personality of the artist.

to be revised, but once you begin to grasp the essence of these exercises, the invention of similar ones to fit new situations will not present a great problem.

The procedure just described should begin each playing session, especially when new material is attempted. The following summary can be used for future reference:

- (1) preparation of the reference sheets before the session.
- (2) study of the material by each student *before* playing.
- (3) reading of the reference sheet while listening to the rhythm section play a few choruses.
- (4) playing of roots in whole notes (or duration of the chords).
- (5) playing of other members of the chords for their durations.
- (6) playing of the chords in various arpeggiated figures.

Since you will feel some self-consciousness and confusion at the first session as to how to begin creating spontaneously, it would be helpful to work on an exercise which will dictate a simple rhythm and which will involve simultaneous improvisation by several students. The first improviser could play a chorus of whole notes, followed by a chorus of half notes, then a chorus of quarter notes, and finally a chorus of any rhythm he chooses, but with eighth notes as the predominant rhythm. (This will also develop an orientation to rhythmic levels, so that you will be aware of whether whole notes, half notes, quarters, etc. are being played.) Choose any pitches, perhaps from the reference sheet at the beginning. Remember to use scale notes as well as chord notes, especially at the quarter-note and eighth-note levels. After the first player has completed his chorus of whole notes, the second begins playing his chorus of whole notes (while the first is playing half notes), then a third person begins playing whole notes, and so on. As each person completes his chorus of eighth notes he drops out until his turn comes around again to start another series of choruses like his first. When a fourth person begins playing, there will be four different rhythmic levels going on simultaneously, which will help relieve self-consciousness. This experiment, charted, would look like:

	Chorus 1		Chorus 2		3	4	5	6	7
1st Player	o →	d →	f →	a →					
2nd Player		o →	d →	f →	a →				
3rd Player			o →	d →	f →	a →			
4th Player				o →	d →	f →	a →		
5th Player					o →	d →	f →	a → etc.	
6th Player						o →	d →	f → etc.	
etc.									

When you all feel comfortable in your playing situation, the preceding exercise can be eliminated in favor of several choruses each, accompanied only by the rhythm section. However, the above chart may be useful later in plotting ways to experiment with simultaneous and contrapuntal improvisation. It would also be advisable to transpose the progression used in this session so you will be able to play in all twelve keys.

This constitutes the rudimentary setting up of the first playing session. When the basic requirements for group playing become rote, then it is possible to bring in added considerations. For example, you should try to become aware of certain pitches within chords which seem more important than others in establishing chords and chord sequences. In examining the progression on page 5, we find that there is a need to clarify the third of the B $\flat$ chord as being major (D) rather than minor (D $\flat$). Similarly, the seventh of the E $\flat$ 7 chord must be D $\flat$ rather than D. The notes D and D $\flat$ play a very important role in establishing the sound of the blues in the key of B $\flat$. The blues progression has the peculiarity of seeming to shift between major and minor, without using a B $\flat$ minor chord. This effect is caused by the E $\flat$ 7 chords, which contain the D $\flat$'s that would have been the most important note of a B $\flat$ minor chord, had there been one. Then the primary function of the E $\flat$ 7 chord is to create a feeling of B $\flat$ minor in the second, fifth, and sixth measures of the blues progression.

Another important pitch in the B $\flat$ blues is the E $\flat$ that occurs the third of the Cm7 chord in the ninth measure, and as seventh of the F7 chord in the tenth measure. This E $\flat$ strengthens the key feeling of B $\flat$, yet it also delays the resolution to the tonic (B $\flat$ M7) chord, which is one of the few notes which cannot be sounded consonantly with a B $\flat$ chord. In conclusion, the most important notes in the blues lie within the small area of a major second from D $\flat$ to E $\flat$, and the charting of these important notes would be:

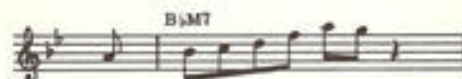
You may now begin to notice that if you play the notes of a chord in rapid succession as pickup notes to a phrase, it becomes easier to hear what should be played. Tonal memory is much like eyesight in that an impression is made which lingers. A tonic registers long enough to retain the sound of a chord for a while after the sound has stopped. When a chord is played in arpeggiated form, the sound is established long enough to be heard simultaneously with subsequent notes. Eventually you will find that you need play only the more important notes of a chord to establish the sound of the entire chord. These notes will usually be the third and the seventh, because their chromatic position determines whether the chord is a M7, m7, or 7, while the fifth and root usually remain stationary.

Notice that in the diagram given above for the Cm7 and the B $\flat$ 7 there are two pitches given as important; the B $\flat$ to A have not been discussed. When either a m7 or a 7 chord resolves to a seventh chord which is located a perfect fourth above (from C to F is a perfect fourth), the most effective way to establish the joining of these chords is to move from the seventh of the first chord (B $\flat$ in this example, since it is the seventh of the Cm7) to the third of the second chord (A, which is the third of an F7). The reason for this is that the other notes of the Cm7 chord—C, E $\flat$, and G—could be

fifth, seventh, and ninth of the F chord and are inconsequential because they are stationary during the change of chords.

If any of the players has difficulty improvising with a particular chord, let the pianist strike the chord and hold it (not in tempo) until the player can improvise a number of useful phrases.

As the beginning improviser finds certain phrases he likes to use from time to time, he is confronted with the problem of fitting the phrase into the meter, so that it starts and ends at the proper time within the measure. For example, suppose you like to play a B $\flat$ M7 in this way:



Now suppose you arrive (in a continuous line) at the junction of your previous idea and this phrase, but a little too early or too late. Always be prepared to cope with these unexpected situations by improvising a few pitches which will enable you to place the idea where it metrically should be. Therefore, instead of turning out like:



it might be:



or:



or:



It is also suggested that in future sessions you prepare some of the motifs from your collection, transposing them and altering them slightly so that they can be used in improvisations at *any point*.

project

Write an original solo on the blues progression, in any key, one chorus in length, using one or two (only) motifs from your collection of motifs. Strive for naturalness of style.

him in an improvising session if he is to fulfill his obligations toward group improvisation, and if he is to learn from listening and *understanding* the efforts of those around him.

sound

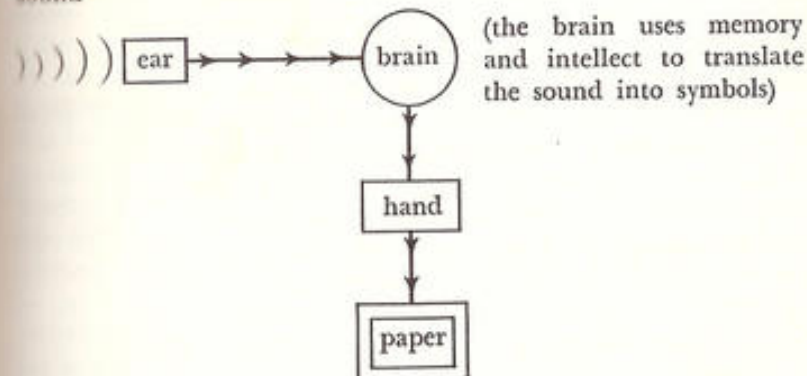


FIGURE 5

Dictation from an Outer Source

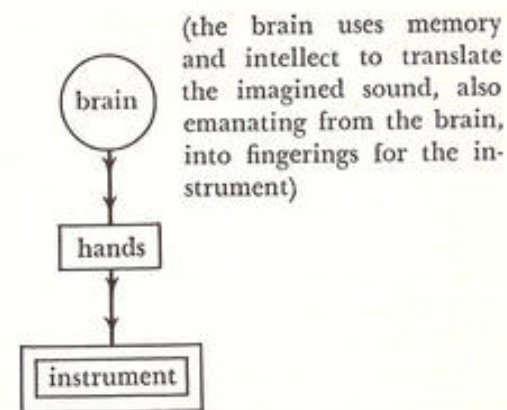


FIGURE 6

Dictation from an Inner Source

5

Development of the Ear

When a student of music theory identifies and writes what his instructor plays for him in a dictation exercise, he is using the same process used by the jazz improviser. He is translating abstract sound into tangible symbols and making it understandable. The student is translating sounds coming to him through the auditory senses and placing these sounds (in symbol form) on paper. The improviser is working with imagined sounds which, when translated, are played on his instrument. The mechanical process of translation, however, is the same and can be developed through extensive practice. Figures 5 and 6 are illustrations of these two types of dictation.

With composers, the process is nearly identical to that of the improviser, except that the brain translates the imagined sounds into symbols to be placed on paper. The important thing to note is that in *any* case, the main process is the same, that of dictation. If the student of jazz can transcribe what he hears from an outside source, he can translate what he hears from an inner source through the same developed technique of taking dictation. Therefore, if he practices transcribing music, regardless of the style, he will greatly increase his ability to improvise. The improviser will often need to use his transcribing ability to translate the sounds around

The memory plays an important part in that it must record imagined sounds instantly so that they can be taken in dictation before the images disappear. Successful efforts can be retained in the memory from one session to another. Improvisation, like composition, is the product of everything heard in past experience, plus the originality of the moment. The contents of even a very accomplished improviser's solos are not all fresh and original, but are a collection of clichés, established patterns, and products of the memory, rearranged in new sequences, along with a few new ideas.

A valuable aid to the improviser is the development of relative pitch on his instrument. This enables the brain to make efficient, accurate translations of heard pitches (from either an inner or outer source) into fingerings necessary to produce that pitch on the instrument. A player who has used his instrument to transcribe melodies (especially if the melody is played on the same type of instrument) has probably discovered that each pitch on his instrument has a slightly different tone color from the others. For example, most pianists can tell when the black keys on a piano are struck in a sequence containing mostly white keys. They can also hear a great difference between pieces played in the key of D $\flat$ and pieces played in the key of E, usually describing D $\flat$ as dark, heavy, and morose, and E as bright, brilliant, and light. String players can often detect a change of strings when listening to another player, especially if the type (gut, metal, wound, and so forth) of string changes within a line. Saxophone, clarinet, and flute players can recognize a difference in quality at the register changes (where the player moves between pitches requiring few or no fingers, to pitches requiring many fingers, though the pitches may be only a minor second interval apart). These minute differences in quality can lead to a type of relative pitch which applies only to the particular instrument played by the performer. It can be developed through dictation exercises (using the instrument, rather than committing the material to paper) and the memorization of the qualities found on various pitches of the instrument. After the improviser develops this sense of pitch to a high degree, he will find that he can identify the exact pitches on different instruments by hearing the note played as it would sound on *his* instrument and identifying it by quality. This is why many saxophone players who switch from

an E $\flat$ -pitched instrument to a B $\flat$ -pitched one are unable to improvise with ease, especially if they use only their ear to guide them. They find themselves playing pitches that are a perfect fourth away from the pitches they intended to produce. Since the fingerings are the same on the new instrument, he has not made allowance for the fact that the two instruments are pitched a perfect fourth apart.

The advantages of acquiring keen pitch perception on a particular instrument are many. First, it practically eliminates all errors involving pitch translation. Second, it enables the player to become proficient in the area of melodic imitation in group situations. Third, the player becomes adept at comprehending the unfamiliar harmonies of new tunes, or different progressions to already learned material. Last, and perhaps most important, the improviser is aware, on the first hearing, of the exact pitches used by other performers.

To develop this most invaluable aid to improvising, it is suggested that you adopt the following methods:

- (1) Begin observing the slight differences in the tone quality of various pitches and registers on your instrument.
- (2) Meet with another player of the same instrument and practice tone-matching exercises, beginning with one pitch at a time and gradually working toward longer series.
- (3) Begin transcribing from recordings of solos of your instrument, playing them on the instrument rather than committing them to paper.

Remember that developing this sense of pitch is a long-term project and that progress may be slow.

6

Further Study of Chord Types

Up to this point, we have used only three types of chord structures, the M7, the m7, and the 7 chords. Frequently, for the sake of variety or to harmonize melody notes which are not within one of the preceding chord structures, alternate chords are used. As will be pointed up in Chapter 10, there are three main functional categories of chords, and the three we have taken up so far are the most commonly used chords of each of the families.

In place of a M7, for example, a *M6 chord* may be substituted. The M6 chord is like the M7 in that it uses a major triad for the bottom three notes, but adds the sixth major scale degree instead of the seventh. A CM6, then, would be spelled C, E, G, A.

If the tune used in an improvisation is in a minor key, then it is necessary to learn a type of minor chord, especially to be used as a tonic minor chord, which differs in sound and construction from the m7 chord. There are two basic examples of this tonic minor sound, the *m6 chord* and a *minor chord with a major seventh* (interval), which we will call a *m#7* or a *m♮7* (depending on whether it is necessary to use a sharp sign or a natural sign to show how the seventh has been raised). A Cm6 chord would be spelled C, E♭, G, A, and the Cm#7 would be spelled C, E♭, G, B♮ (not B♭, as in the Cm7).

Sometimes a m7 chord may be replaced by a *half-diminished seventh chord* (ø7) which differs from the minor seventh chord only in that the fifth is lowered a half step. A Cø7 chord, instead of being spelled C, E♭, G, B♭ (Cm7), is spelled C, E♭, G♭, B♭.

A C♭5 or a C+5 may often be used in place of a C7; they are spelled C, E, G♭, B♭, and C, E, G♯, B♭ respectively, rather than C, E, G, B♭.

All the newly given alternate chords may be used in place of their given and more common chord family member (M7, m7, or 7) quite freely, sometimes depending upon the harmonization of a given melody. A summary of the families is given below.

M7	'Tonic Minor'	m7	7
M7	m6	m7	7
M6	m#7 or m♮7	ø7	7
			+5
			7
			♭5

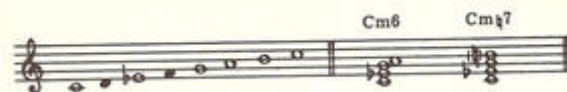
Figure 7 shows a summary of the symbol, name, intervallic construction, and an example of each of the chord types included thus far.

Most of the newly added chords, though they belong to families of already learned chords, will require scales different from those given in Chapter 1 for the M7, m7, and 7 chords. It will be remembered that a major scale was used for the M7 chord, a Dorian Mode for m7 chords, and the Mixolydian Mode for the 7 chord, the constructions of which are given in Chapter 1.

The M6 chord will use the same scale as the M7 chord. However, the m6 and m#7 (or m♮7), depicting the minor mode, will differ from either the M7 or m7 in respect to the scale used with it. The ascending form of the melodic minor scale (see p. 40) may be used with either the m6 or the m#7 chords, as the scale contains all the notes of those chords. The harmonic minor scale (see p. 40) may accompany the m#7 chord only, since the lowered sixth degree would conflict with the sixth of the m6 chord.

SYMBOL	NAME	INTERVALS CONTAINED	EXAMPLE
M7	major seventh chord	major third perfect fifth major seventh	CM7
M6	major sixth chord	major third perfect fifth major sixth	CM6
m6	minor sixth chord	minor third perfect fifth major sixth	Cm6
m $\sharp$ 7 or m $\natural$ 7	minor chord with a major seventh	minor third perfect fifth major seventh	Cm $\sharp$ 7
m7	minor seventh chord	minor third perfect fifth minor seventh	Cm7
ϕ 7	half-diminished seventh chord	minor third diminished fifth minor seventh	C ϕ 7
7	seventh chord or dominant seventh chord	major third perfect fifth minor seventh	C7
$\overset{7}{+5}$ or $+7$	augmented seventh chord	major third augmented fifth minor seventh	C $+7$
$\overset{7}{\flat 5}$	seventh chord with a diminished fifth	major third diminished fifth minor seventh	C $\overset{7}{\flat 5}$

FIGURE 7



C melodic minor scale (ascending)



C harmonic minor scale

The scale which best fits the ϕ 7 chord is one which uses the notes of a major scale of a half step up (called Locrian mode), hence a scale on B would use the notes of a C major scale, but starting on B (B, C, D, E, F, G, A, B). This type of scale contains all the notes of a ϕ 7 chord built on the root of the scale.

Locrian Mode on C (D $\flat$ major scale, starting on C)

One appropriate scale fits both the $\overset{7}{+5}$ and $\overset{7}{\flat 5}$ chords—the *whole-tone scale*, which, as its title suggests, is constructed by using successive whole steps *only*.



C whole-tone scale